We summarize the results:

Trapezoidal Rule

$$\int_a^b f(x)\,dx = \frac{h}{2}[f_0 + 2f_1 + 2f_2 + \cdots + 2f_{n-1} + f_n] - \overbrace{\frac{h^3}{12}f''(\xi)}^{\text{local error}}$$

$$\text{Global error} = -\frac{(b-a)}{12}h^2 f''(\xi).$$

Simpson's $\frac{1}{3}$ Rule

$$\int_a^b f(x)\,dx = \frac{h}{3}[f_0 + 4f_1 + 2f_2 + 4f_3 + \cdots + 2f_{n-2} + 4f_{n-1} + f_n] - \overbrace{\frac{h^5}{90}f^{iv}(\xi)}^{\text{local error}}$$

$$\text{Global error} = -\frac{(b-a)}{180}h^4 f^{iv}(\xi).$$

Rule for Milne's Method

$$\int_{x_{n-3}}^{x_{n+1}} f(x)\,dx = \frac{4h}{3}[2f_{n-2} - f_{n-1} + 2f_n] + \overbrace{\tfrac{28}{90}h^5 f^{iv}(\xi)}^{\text{local error}}.$$

Rules for Adam's Method

$$\int_{x_n}^{x_{n+1}} f(x)\,dx = \frac{h}{24}[-9f_{n-3} + 37f_{n-2} - 59f_{n-1} + 55f_n] + \overbrace{\tfrac{251}{720}h^5 f^{iv}(\xi)}^{\text{local error}},$$

$$\int_{x_n}^{x_{n+1}} f(x)\,dx = \frac{h}{24}[f_{n-2} - 5f_{n-1} + 19f_n + 9f_{n+1}] - \overbrace{\tfrac{19}{720}h^5 f^{iv}(\xi)}^{\text{local error}}.$$

PROBLEMS

Section 1

1. Derive formulas similar to Eqs. (1.4) and (1.6) for $f'(x_0)$, the derivative at x_0, but in terms of f-values to the left of x_0, e.g., evaluate the constants in the relations $f_0' = c_0 f_0 + c_{-1} f_{-1}$ and $f_0' = c_0 f_0 + c_{-1} f_{-1} + c_{-2} f_{-2}$. [Take the x-values as equally spaced, and $\Delta x = h$.]

2. When more information about the function is used to compute the derivative, a more accurate formula results. Hence the formula $f_0' = c_0 f_0 + c_1 f_1 + c_2 f_2 + c_3 f_3$ is more accurate than either Eq. (1.4) or (1.6). Find the constants assuming equispaced x-values.

3. When values on both sides are used in a derivative formula, the accuracy is improved over a one-sided formula. Derive a formula more accurate than the four-term for-